

Real Numbers.

1) Euclid's division lemma and its application

general formula $\rightarrow a = bq + r, 0 \leq r < b$.

2) Euclid's division algorithm,

dividend = divisor $\times$ quotient + remainder.

3) Product of 2 no.s = HCF $\times$ LCM

$$* 1^{\text{st}} \text{ no.} = \frac{\text{HCF} \times \text{LCM}}{2^{\text{nd}} \text{ no.}}$$

$$* 2^{\text{nd}} \text{ no.} = \frac{\text{HCF} \times \text{LCM}}{1^{\text{st}} \text{ no.}}$$

$$* \text{HCF} = \frac{\text{product of two no.s}}{\text{LCM}}$$

$$* \text{LCM} = \frac{\text{product of 2 given no.s}}{\text{HCF}}$$

4) Determining the nature of decimal expansion (terminating or non-terminating, recurring) of rational no.s by the formula $2^n 5^m$.

5) Proving irrationality of numbers $[\sqrt{3}-\sqrt{2}, \frac{1}{\sqrt{5}}+6, \frac{1}{\sqrt{2}}, \sqrt{10}]$

Polynomials

1) Linear polynomial $\rightarrow$ degree 1

Quadratic polynomial $\rightarrow$ degree 2

Cubic polynomial $\rightarrow$ degree 3.

2) the zero of the linear polynomial $ax+b$ is $-\frac{b}{a} = \frac{\text{constant term}}{\text{coefficient of } x}$.

3) general formula of Quadratic Polynomial $\rightarrow ax^2+bx+c$

4) Relationship between zeroes and co-efficient of a polynomial.

$$* \text{sum of zeroes} = -\frac{b}{a} (\alpha + \beta)$$

$$* \text{product of zeroes} = \frac{c}{a} (\alpha\beta)$$

5) Required quadratic polynomial = $x^2 - (\text{sum of zeroes})x + \text{pdt. of zeroes}$.

6) Division algorithm for polynomials, i.e.,

$$p(x) = q(x) \times a(x) + r(x)$$

$$q(x) = \frac{p(x) - r(x)}{a(x)}$$

4) Applications of zeroes of a polynomial.

$$* \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$* \alpha^2 - \beta^2 = \alpha + \beta \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$$

$$* (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

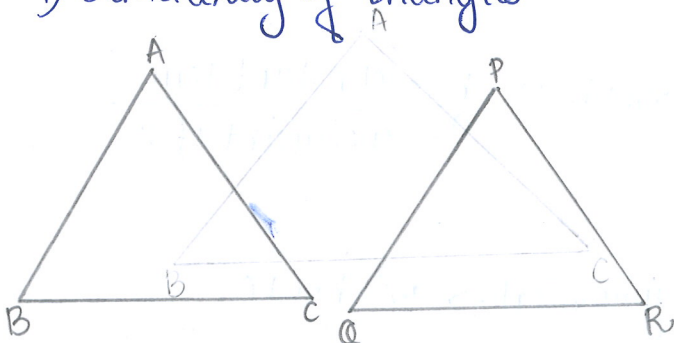
$$* \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$* \alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2)$$

$$= \left[\sqrt{(\alpha + \beta)^2 - 4\alpha\beta} \right] \left[(\alpha + \beta)^2 - \alpha\beta \right]$$

Triangles

1) Similarity of triangles



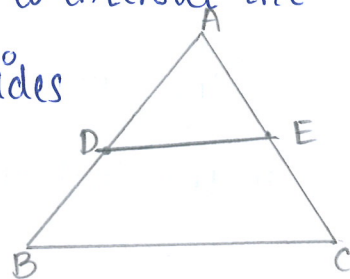
If $\triangle ABC \sim \triangle PQR$

$\angle A = \angle P$; $\angle B = \angle Q$; $\angle C = \angle R$ [corresponding angles are equal]

$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$ [ratio of their corresponding sides are equal].

2) Basic Proportionality Theorem

- If a line drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. i.e. $\frac{AD}{DB} = \frac{AE}{EC}$

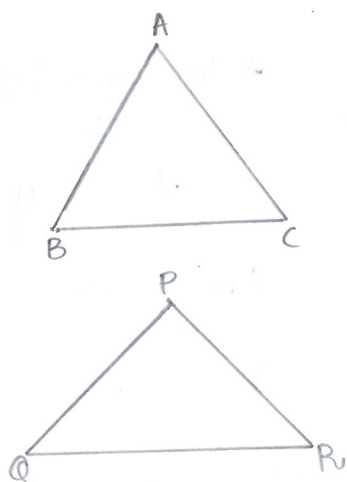


3) Criteria of similarity of triangles

- * AAA (Angle, Angle Angle) criterion. also referred to as AA similarity.
- * SSS (side side side) criteria.
- * SAS (side angle side) criteria.

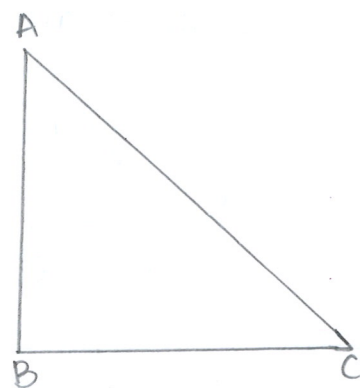
4) Area Theorem

The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. i.e. $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2$



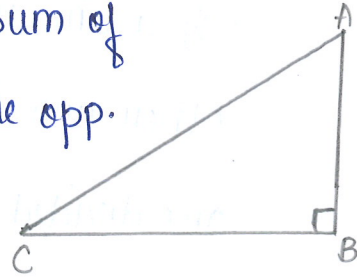
5) Pythagoras Theorem,

If a $\perp$ is drawn from the vertex of the right angle of a right ~~angle~~ triangle to the hypotenuse, then triangles on both sides of a $\perp$ are similar to the whole triangle and to each other. OR In a right Δ , the square of the hypotenuse is equal to the sum of squares of the other two sides. i.e., $AC^2 = AB^2 + BC^2$



6) Converse of Pythagoras Theorem

In a Δ , if the square of one side is equal to the sum of the squares of the other two sides, then the angle opp. to the first side is a right angle. i.e., $\angle B = 90^\circ$



Pair of Linear Eqn. in Two Variables

1) General formula of Linear Eqn. $\rightarrow a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$

2) Solving x and $y \rightarrow$ Graphical method and Algebraic method.

3) * Elimination method

* Substitution method

* Cross multiplication

$\downarrow$

x	y	$= 1$
b_1	c_1	a_1
b_2	c_2	a_2

(Note: In the original image, lines connect b1 to c2, c1 to a2, and a1 to b2, forming a cycle for cross-multiplication.)

* When time increases, speed decreases
* When speed increases, time decreases.

4) $\frac{5}{x-1} + \frac{1}{y-2} = 2$; $\frac{6}{x-1} - \frac{3}{y-2} = 1$

In this case, let us put $\frac{1}{x-1} = p$ and $\frac{1}{y-2} = q$ and solve it.

5) There are 3 different solns. which can arise when a pair of linear equation is given:-

(i) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: In this situation, linear pair is consistent, lines intersect at a point and has a unique solution.

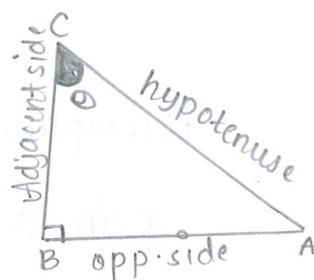
(ii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: In this case, the eqn. is inconsistent, has no solns and the lines are parallel.

(iii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$: In this case, the eqn. is dependent and consistent, has infinitely many solutions and lines are coincident.

6) Speed = $\frac{\text{distance}}{\text{time}}$; time = $\frac{\text{distance}}{\text{speed}}$; distance = speed $\times$ time.

Introduction to Trigonometry.

1) The trigonometric ratios of the $\angle C$ in right $\triangle ABC$ is,



$$* \sin C = \frac{\text{opp. side}}{\text{adjacent side}} = \frac{\text{Opposite side}}{\text{Hypotenuse}} = \frac{O}{H} = \frac{BA}{CA}$$

$$* \cos C = \frac{\text{adjacent side}}{\text{hypotenuse}} = \frac{A}{H} = \frac{BC}{AC}$$

$$* \tan C = \frac{\text{opposite side}}{\text{adjacent side}} = \frac{O}{A} = \frac{AB}{BC}$$

$$* \text{Cosecant } C = \frac{\text{hypotenuse}}{\text{opposite side}} = \frac{H}{O} = \frac{AC}{AB}$$

$$* \sec C = \frac{\text{hypotenuse}}{\text{adjacent side}} = \frac{H}{A} = \frac{AC}{BC}$$

$$* \cot C = \frac{\text{adjacent side}}{\text{opposite side}} = \frac{A}{O} = \frac{BC}{AB}$$

2) Reciprocal Relations

$$* \sin \theta = \frac{1}{\text{cosec } \theta} \quad * \cos \theta = \frac{1}{\sec \theta} \quad * \tan \theta = \frac{1}{\cot \theta}$$

$$* \text{cosec } \theta = \frac{1}{\sin \theta} \quad * \sec \theta = \frac{1}{\cos \theta} \quad * \cot \theta = \frac{1}{\tan \theta}$$

3) Product Relations

$$* \sin \theta \times \operatorname{cosec} \theta = 1 \quad * \cos \theta \times \sec \theta = 1 \quad * \tan \theta \times \cot \theta = 1$$

$$* \operatorname{cosec} \theta \times \sin \theta = 1 \quad * \sec \theta \times \cos \theta = 1 \quad * \cot \theta \times \tan \theta = 1$$

4) Quotient Relations

$$\sin \theta = \frac{O}{H}; \quad \cos \theta = \frac{A}{H}$$

$$\cos \theta = \frac{A}{H}; \quad \sin \theta = \frac{O}{H}$$

$$* \frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$* \frac{\cos \theta}{\sin \theta} = \cot \theta$$

5) Complementary Relations

$$* \sin (90^\circ - \theta) = \cos \theta \quad * \operatorname{cosec} (90^\circ - \theta) = \sec \theta \quad * \tan (90^\circ - \theta) = \cot \theta$$

$$* \cos (90^\circ - \theta) = \sin \theta \quad * \sec (90^\circ - \theta) = \operatorname{cosec} \theta \quad * \cot (90^\circ - \theta) = \tan \theta$$

6) Trigonometric Identities

$$* \sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta; \quad \cos^2 \theta = 1 - \sin^2 \theta$$

$$* \sec^2 \theta - \tan^2 \theta = 1$$

$$\sec^2 \theta = 1 + \tan^2 \theta; \quad \tan^2 \theta = \sec^2 \theta - 1$$

$$* \operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta; \quad \cot^2 \theta = \operatorname{cosec}^2 \theta - 1$$

Some Applications of Trigonometry.

1) Trigonometric ratios of some specific angles.

$\angle A$	0°	30°	45°	60°	90°
$\sin A$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos A$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan A$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined
$\operatorname{cosec} A$	not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec A$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	not defined
$\cot A$	not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

SOH CAH TOA

Statistics

1) Mean of grouped data

(a) Direct method

$$\text{Mean } \bar{x} = \frac{\sum x_i f_i}{\sum f_i}$$

In a class interval of [25-50]

* lower limit = 25 * upper limit = 50 * class size (h) = 50 - 25 = 25

$$\text{* class mark (x)} = \frac{\text{upper limit} + \text{lower limit}}{2}$$

(b) Assumed Mean Method

$$\text{Mean } \bar{x} = A + \frac{\sum f_i d_i}{\sum f_i}$$

* A = assumed mean * $d_i = x_i - A$

(c) Step deviation method

$$\text{Mean } \bar{x} = A + \left[\frac{\sum f_i u_i}{\sum f_i} \right] \times h$$

* h = class size * $u_i = \frac{x_i - A}{h}$ * A = assumed mean

2) Mode of grouped Data

$$\text{Mode} = L + \left[\frac{f_i - f_0}{2f_i - f_0 - f_2} \right] \times h$$

* L = lower limit of modal class.

* h = size of class interval.

* f_i - frequency of modal class.

* f_0 - frequency of class preceding the modal class.

* f_2 - frequency of class ~~pre~~^{su}ceeding the modal class.

3) Median of grouped Data:

$$\text{Median} = L + \left[\frac{\frac{n}{2} - cf}{f} \right] \times h$$

* L = lower limit of median class.

* n = no. of observations.

* cf = cumulative frequency of class preceding the median class.

* f = frequency of median class.

* h = class size.

To find the median class, find the cf of all classes and $\frac{n}{2}$. We now locate the class whose cf is greater than or nearest to $\frac{n}{2}$.

4) Empirical relationship between the 3 measures of central tendency

$$(\text{mean}, \text{median}, \text{mode}) = 3 \text{ me} = \text{mode} + 2 \text{ mean}.$$

5) Two methods to construct ogives :-

(a) Less than ogive method $\rightarrow$ ogive is cumulated upwards

(b) More than ogive method $\rightarrow$ ogive is cumulated downwards.

Quadratic Equations

1) General formula $\rightarrow ax^2 + bx + c = 0, a \neq 0$

2) * Factorisation method

* Completing the square method

* Quadratic formula

$$\downarrow$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

} solving applications

3) Discriminant $= D = b^2 - 4ac$

4) Nature of roots

* If $b^2 - 4ac > 0$, two distinct real roots.

* If $b^2 - 4ac = 0$, two equal roots.

* If $b^2 - 4ac < 0$, no real roots.

5) upstream is always negative $(x-y)$, downstream is always positive $(x+y)$

6) Two digit no. $= 10x + y$

7) To change km/hr to m/sec, multiply by $\frac{5}{18}$ and m/sec to km/hr multiply by $\frac{18}{5}$.

Arithmetic Progressions

1) General formula of an A.P is $a, a+d, a+2d, a+3d, \dots$

$$2) d = a_2 - a_1 ; a_3 - a_2 ; a_{10} - a_9$$

$$3) a_n = a_1 + (n-1)d$$

$$4) a, b, c \text{ are three consecutive terms then } b = \frac{a+c}{2}$$

$$5) S_1 = a_1 ; S_2 = a_1 + a_2 ; S_2 = a_1 + a_2$$

$$6) S_n = \frac{n}{2} [a_1 + a_n] ; S_n = \frac{n}{2} [2a_1 + (n-1)d]$$

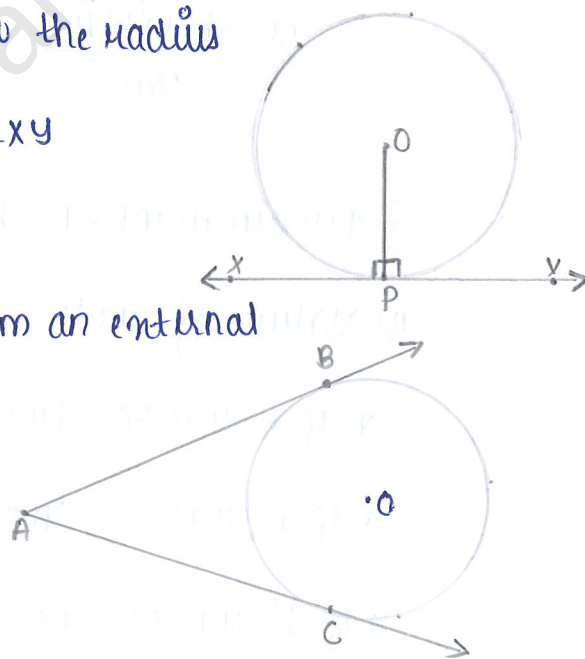
Circles

1) Theorem 10.1

A tangent to a circle is perpendicular to the radius through the point of contact. i.e. $OP \perp XY$

2) Theorem 10.2

The length of two tangents drawn from an external point to a circle are equal. i.e. $AB = AC$



Co-Ordinate geometry

$$1) \text{ Distance formula} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2) In order to prove the given figure is a

- * square, P.T, four sides are equal and diagonals are also equal.
- * parallelogram, P.T, opposite sides are equal, diagonals are unequal.
- * rectangle, P.T, opposite side and diagonals are equal.
- * rhombus, P.T, all sides are equal and diagonals are unequal.

$$3) \text{ Section formula} = \left[\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right]$$

4) point on x-axis = $p(x, 0)$, point of y-axis = $p(0, y)$

5) midpoint formula = $\left[\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right]$

6) Area of a $\Delta = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$

7) Centroid of a $\Delta = \left[\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right]$

8) $|-10| = 10$; $|-4| = 4$

Probability

1) Probability always lies between $0 \leq P(E) \leq 1$.

2) $P(\bar{E}) = 1 - P(E)$

3) $P(E) + P(\bar{E}) = 1$

4) *sure event = 1 *impossible event = 0

5) $P(E) = \frac{\text{no. of outcomes favourable to } E}{\text{total no. of possible outcomes}}$

6) Prime no.s $\rightarrow 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, \dots$

7) Composite no.s $\rightarrow 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39, \dots$

8) Square no.s $\rightarrow 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, \dots$

9) Cubic no.s $\rightarrow 1, 8, 27, 64, 125, 216, \dots$

10) Tossing a coin - Head and Tail (outcomes-2)

ii) Tossing two coins - (outcomes-4)

	H	T
H	HH	HT
T	HT	TT

12) Tossing 3 coins - (Outcomes - 8)

	H	T
HH	HHH	HHT
HT	HTH	HTT
TH	THH	THT
TT	TTH	TTT

13) Throwing a die - (Outcomes - 6)

14) Throwing two dice - (Outcomes - 36)

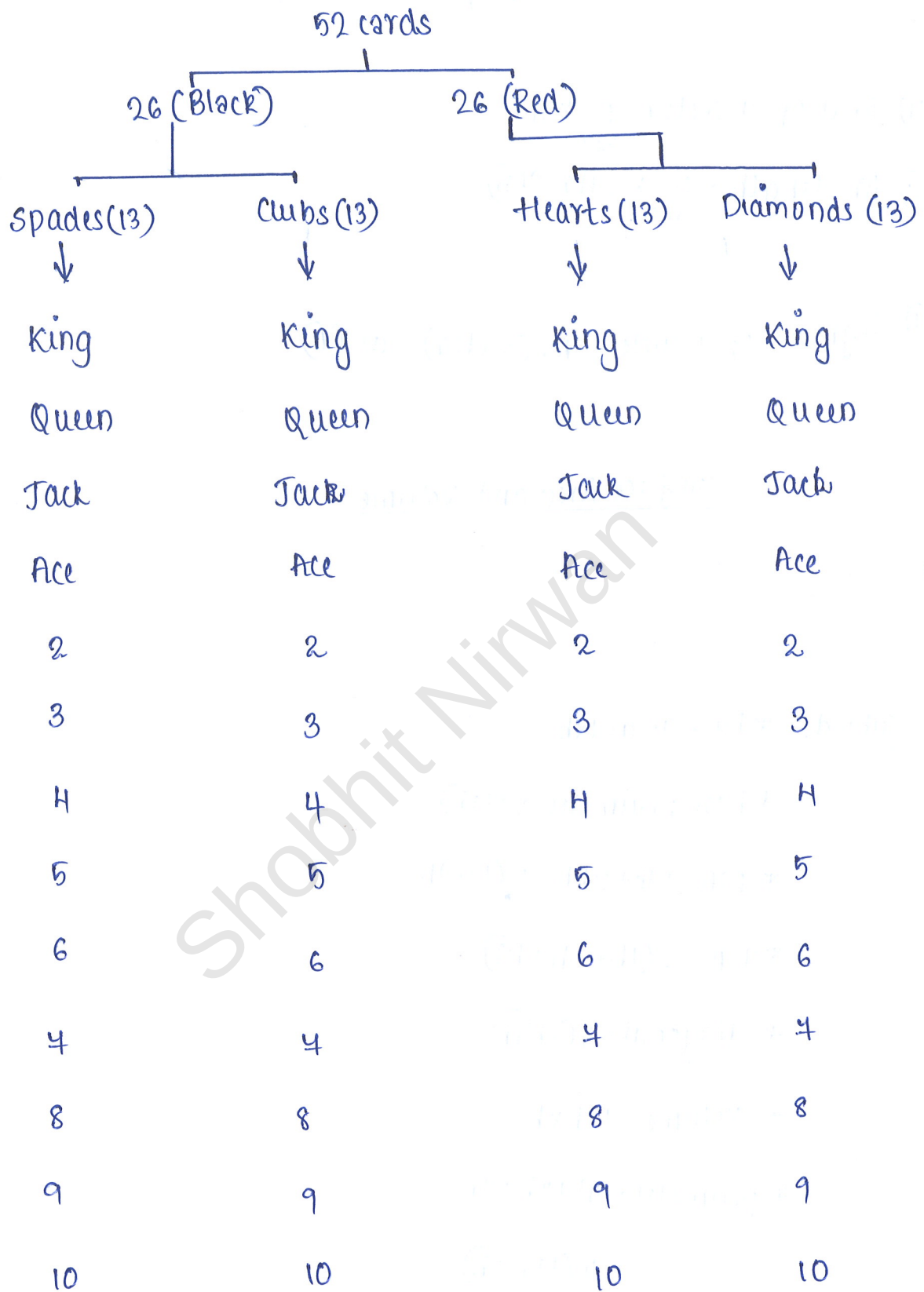
	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

15) Face Cards → *King

*Queen

*Jack

16) Playing Cards



Areas Related to Circles

1) Area of a circle = πr^2 , $d = 2r$

2) Circumference of a circle = $2\pi r$

3) Area of a semi-circle = $\frac{\pi r^2}{2}$

4) Circumference of a semi-circle = $\pi r + 2r$

Area (semicircle) = $\frac{1}{2} \pi r^2$

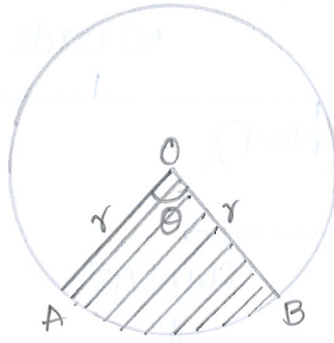
5) Area of a quadrant = $\frac{\pi r^2}{4}$

Circumference (quadrant) = $\frac{\pi r}{2} + 2r$

6) no. of revolutions = $\frac{\text{distance covered}}{2\pi r}$

4) Area of a sector = $\frac{\theta}{360} \times \pi r^2$

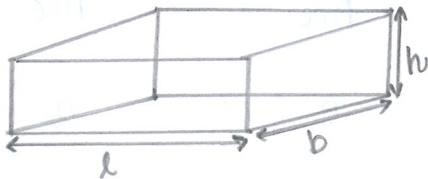
8) Arc Length = $\frac{\theta}{360} \times 2\pi r$ (AB)



9) Segment of a circle = Ar (sector) - ar (Δ)

Surface Area and Volume

1)



Cuboid:- * base area = $l \times b$

* base perimeter = $2(l+b)$

* LSA = $2lh + 2bh = 2(l+b)h$

* TSA = $2(lb + bh + h^2)$

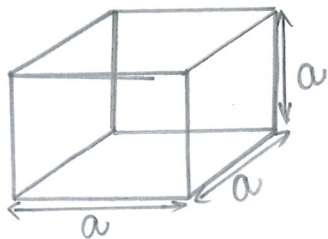
* diagonal = $\sqrt{l^2 + b^2 + h^2}$

* Volume = $l \times b \times h$

* perimeter = $4l + 4b + 4h$

= $4(l+b+h)$

2)



Cube:- * base area = a^2

* LSA = $4a^2$

* TSA = $6a^2$

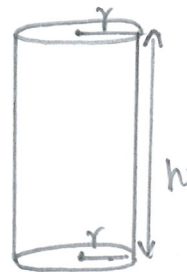
* $V = a^3$

* base perimeter = $4a$

* perimeter = $12a$

* diagonal = $\sqrt{a^2 + a^2 + a^2} = \sqrt{3}a$

3)



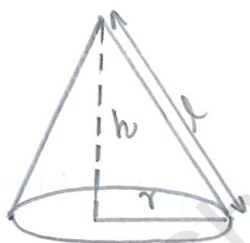
Cylinder:- * base area = πr^2

* CSA = $2\pi r h$

* TSA = $2\pi r (r + h)$

* $V = \pi r^2 h$

4)



Cone:- * base area = πr^2

* CSA = $\pi r l$

* TSA = $\pi r (r + l)$

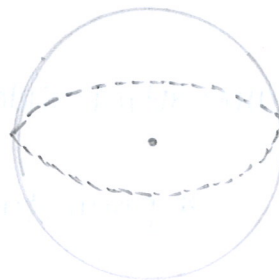
* $V = \frac{1}{3} \pi r^2 h$

* $l = \sqrt{h^2 + r^2}$

* $h = \sqrt{l^2 - r^2}$

* $r = \sqrt{l^2 - h^2}$

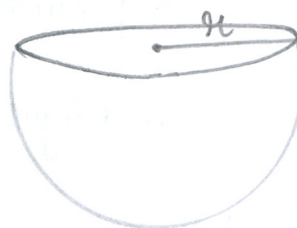
5)



Sphere:- * CSA = TSA = $\frac{4}{3} \pi r^2$

* $V = \frac{4}{3} \pi r^3$

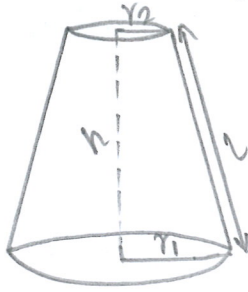
6)



Hemisphere:- * base area = πr^2

* CSA = $2\pi r^2$ * TSA = $3\pi r^2$ * $V = \frac{2}{3} \pi r^3$

4)



Frustum of a cone:- * base area = πr^2

$$* \text{CSA} = \pi l (r_1 + r_2)$$

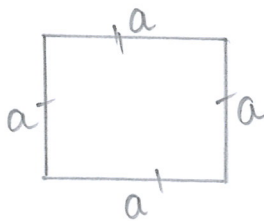
$$* l = \sqrt{h^2 + (r_1 - r_2)^2}$$

$$* \text{TSA} = \pi r_1^2 + \pi r_2^2 + \pi l (r_1 + r_2)$$

$$= \pi (r_1^2 + r_2^2) + \pi l (r_1 + r_2)$$

Things to Remember:-

1)

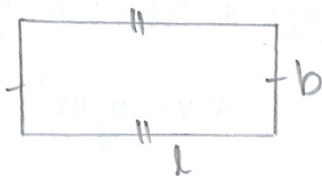


Square:- * area = side $\times$ side

$$* \text{perimeter} = 4 \times \text{side}$$

$$* \text{diagonal} = \sqrt{2} a$$

2)



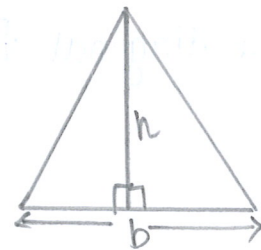
Rectangle:- * area = $l \times b$

$$* \text{perimeter} = 2(l + b)$$

$$* \text{diagonal} = \sqrt{l^2 + b^2}$$

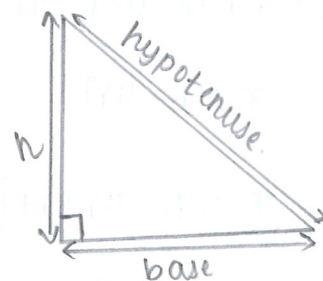
$$3) \pi = 3.14 \mid \pi = \frac{22}{7}$$

4)



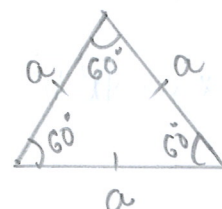
Triangle:- * Area = $\frac{1}{2} \times b \times h$

5)



Right Δ :- Area = $\frac{1}{2} \times b \times h$

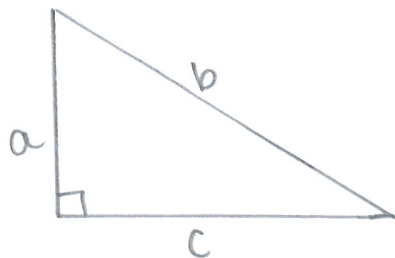
6)



Equilateral Δ :- * Area = $\frac{\sqrt{3}}{4} a^2$

$$* \sqrt{3} = 1.43$$

4)



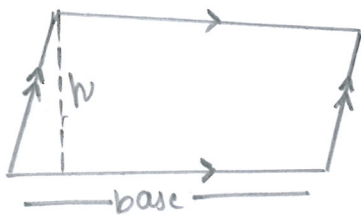
Pythagoras theorem:-

$$* b^2 = a^2 + c^2$$

$$* a^2 = b^2 - c^2$$

$$* c^2 = b^2 - a^2$$

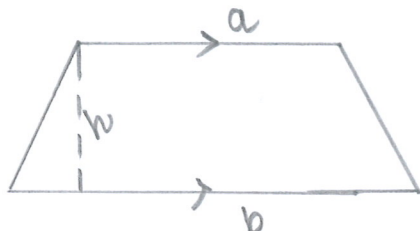
8)



Parallelogram:-

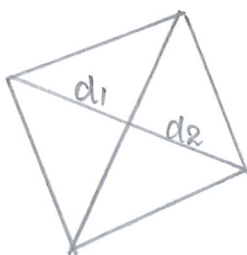
$$\text{Area} = b \times h.$$

9)



$$\text{Trapezium:- Area} = \frac{1}{2} (a+b)h$$

10)



$$\text{Rhombus:- Area} = \frac{1}{2} \times d_1 \times d_2$$

11) Complementary angles $\rightarrow$ sum of two angle is 90° .

12) Supplementary angles $\rightarrow$ sum of 2 angle is 180° .

13) Collinear points $\rightarrow$ Points which are on the same straight line. Area = 0

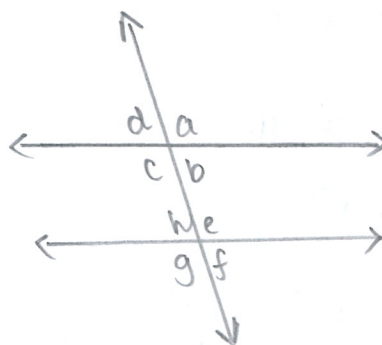
14) If two parallel lines intersects by a transversal,

$$\left. \begin{array}{l} \angle b = \angle h \\ \angle c = \angle e \end{array} \right\} \text{alternate interior angles.}$$

$$\left. \begin{array}{l} \angle d = \angle h \\ \angle c = \angle g \\ \angle a = \angle e \\ \angle b = \angle f \end{array} \right\} \text{corresponding angles.}$$

$$\angle c + \angle h = 180^\circ$$

$$\angle b + \angle e = 180^\circ$$



15) Identities:-

$$* (a+b)^2 = a^2 + b^2 + 2ab$$

$$* (a-b)^2 = a^2 - 2ab + b^2$$

$$* (a+b)(a-b) = a^2 - b^2$$

$$* (x+a)(x+b) = x^2 + (a+b)x + ab$$

$$* (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$* (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$* a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$* a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$* (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$$

$$* x^2 + y^2 = (x+y)^2 - 2xy$$

$$* x^2 + y^2 = (x-y)^2 + 2xy$$

$$* (a^m)^n = a^{m \times n}$$

$$* a^m \times a^n = a^{m+n}$$

$$* a^m \div a^n = a^{m-n}$$

$$* a^0 = 1$$

$$* a^1 = a$$

$$* a^{-1} = \frac{1}{a}$$

$$* \left(\frac{a}{b}\right)^{-2} = \left(\frac{b}{a}\right)^2$$

$$* \sqrt{a} = a^{1/2}$$

$$* \sqrt{a} \times \sqrt{a} = a$$